

Clinical Motivation

Early atrophy (EA) is a precursor to advanced atrophic change in age-related macular degeneration. Clinicians have also observed a vertical hypertransmission pattern in baseline OCT, informally called **barcoding**, which may indicate regions at increased risk of progression.

A scalable system should localize both findings while showing clinicians which image regions influenced each decision. This poster proposes a two-output segmentation network with built-in visual explanation.

Input and Preprocessing

Each example contains an OCT B-scan and separate manual masks for **EA** and **barcoding**. Anatomical preprocessing standardizes the network inputs:

1. Load the B-scan and Bruch membrane (BM) boundary.
2. Flatten the scan to BM and retain the sub-BM region.
3. Normalize intensity within the scan.
4. Apply mild denoising while preserving vertical patterns.
5. Resize or pad every image and mask to a common tensor shape.
6. Apply paired, label-preserving augmentation during training: horizontal reflection, small translation, crop, and intensity jitter.

Split by patient *before* augmentation to prevent data leakage. A validity mask V removes uncertain and vessel/structural annotations from the loss rather than incorrectly treating those pixels as normal tissue.

What the Network Predicts

For every image location i , the model returns two probabilities:

$$p_i^{EA} = P(y_i^{EA} = 1 \mid I), \quad p_i^{BC} = P(y_i^{BC} = 1 \mid I).$$

Separate heads share anatomical features without sharing a decision boundary.

With phenotype-specific threshold τ_k , the binary prediction is $\hat{y}_i^k = \mathbb{I}[p_i^k \geq \tau_k]$. Small isolated components can be removed using a minimum-width rule selected only on validation data.

Saved outputs include both probability maps and masks; interval count, width, location, and total burden; phenotype-specific Grad-CAM maps; and an overlay linking predictions to the original B-scan.

Why Two Segmentation Heads?

EA and barcoding share retinal boundaries, brightness, and hypertransmission, but their spatial criteria differ. A shared decoder learns common context; separate 1×1 convolution heads learn phenotype-specific maps. The masks remain non-exclusive unless clinical labeling establishes another rule. Each task may use distinct class weights, loss weights, thresholds, and calibration curves.

Proposed ResNet–U-Net Architecture

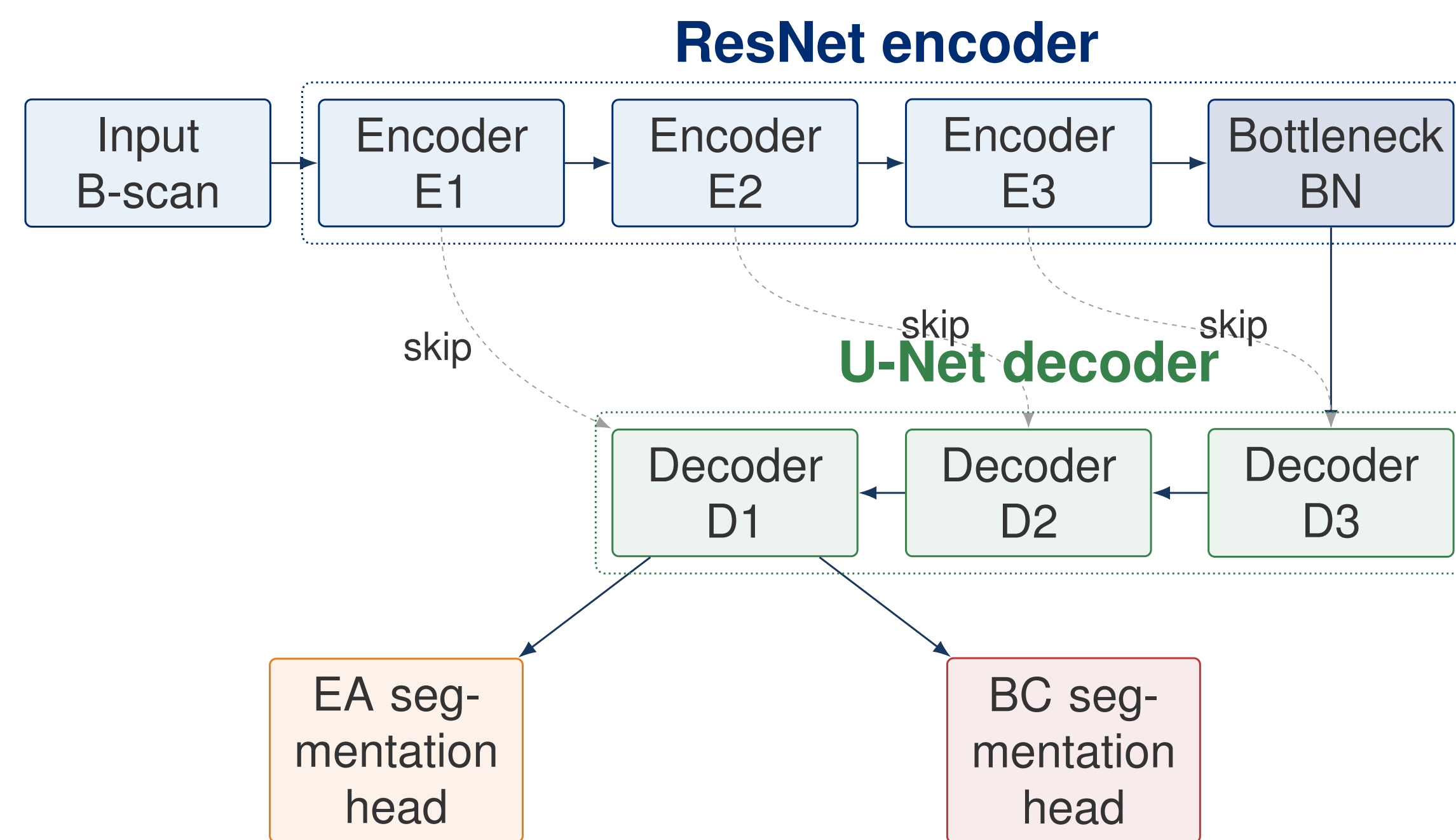


Diagram key. E1 captures edges and fine texture; E2 captures local retinal anatomy; E3 captures deeper spatial context; BN is the compressed bottleneck representation. D3, D2, and D1 successively upsample and fuse encoder features. BC denotes barcoding. Each segmentation head applies a 1×1 convolution and sigmoid to produce its probability map.

The ResNet encoder compresses the scan into increasingly contextual features. Residual connections make deeper optimization stable. The U-Net decoder restores spatial resolution, while skip connections recover fine retinal boundaries that would otherwise be lost during downsampling.

Shared Representation, Separate Decisions

Let $f_\theta(I)$ be the shared encoder–decoder representation. The two output heads are

$$\hat{Y}_{EA} = \sigma(W_{EA} * f_\theta(I) + b_{EA}), \quad \hat{Y}_{BC} = \sigma(W_{BC} * f_\theta(I) + b_{BC}).$$

Sharing most parameters reduces the amount of data needed relative to two completely independent networks. Separate heads preserve phenotype-specific calibration and allow different probability thresholds after training.

From Masks to Quantitative Biomarkers

Connected predicted columns are converted into intervals. For phenotype $k \in \{EA, BC\}$:

$$N_k = \text{number of intervals}, \quad B_k = \sum_{r=1}^{N_k} w_{k,r},$$

where $w_{k,r}$ is interval width and B_k is total horizontal burden. Reporting probability maps alongside these summaries preserves information for later threshold calibration and statistical analysis.

Training Objective

Each head combines weighted binary cross-entropy, which addresses class imbalance, with Dice loss, which rewards spatial overlap:

$$\mathcal{L}_k = \alpha_k \mathcal{L}_{WBCE}^k + (1 - \alpha_k) \mathcal{L}_{Dice}^k.$$

The joint objective is

$$\mathcal{L} = \lambda_{EA} \mathcal{L}_{EA} + \lambda_{BC} \mathcal{L}_{BC}.$$

The weights are chosen on the validation set so the rarer or more difficult phenotype does not become dominated by the other task.

Proposed Training Algorithm

Algorithm 1 Patient-level multi-task segmentation training

Require: Patient dataset $\mathcal{D} = \{(I_n, Y_n^{EA}, Y_n^{BC}, V_n)\}_{n=1}^N$; epochs T ; learning rate η ; loss weights $\lambda_{EA}, \lambda_{BC}$

Ensure: Parameters θ^* and thresholds $(\tau_{EA}^*, \tau_{BC}^*)$

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1: Split patient IDs into  $\mathcal{D}_{tr}, \mathcal{D}_{val}, \mathcal{D}_{te}$ 
2: Initialize ResNet encoder  $E_{\theta_E}$  from pretrained weights; initialize decoder  $D_{\theta_D}$  and heads  $h_{EA}, h_{BC}$ 
3: for  $t = 1, \dots, T$  do
4:   for mini-batch  $\mathcal{B} \subset \mathcal{D}_{tr}$  do
5:      $(\tilde{I}, \tilde{Y}^{EA}, \tilde{Y}^{BC}, \tilde{V}) \leftarrow \text{Augment}(\mathcal{B})$ 
6:      $F \leftarrow D_{\theta_D}(E_{\theta_E}(\tilde{I}))$ 
7:     for  $k \in \{EA, BC\}$  do
8:        $P^k \leftarrow \sigma(h_k(F))$ 
9:        $\mathcal{L}_k \leftarrow \alpha_k \mathcal{L}_{WBCE}(P^k, \tilde{Y}^k; \tilde{V}) + (1 - \alpha_k) \mathcal{L}_{Dice}(P^k, \tilde{Y}^k; \tilde{V})$ 
10:    end for
11:     $\mathcal{L} \leftarrow \lambda_{EA} \mathcal{L}_{EA} + \lambda_{BC} \mathcal{L}_{BC}$ 
12:     $g \leftarrow \nabla_{\theta} \mathcal{L}$ ;  $\theta \leftarrow \text{AdamW}(\theta, g, \eta)$ 
13:  end for
14: Evaluate  $\mathcal{L}_{val}^{EA}$  and  $\mathcal{L}_{val}^{BC}$ ; update scheduler
15: if joint validation loss improves then
16:    $\theta^* \leftarrow \theta$ ; save checkpoint
17: else if patience is exhausted then
18:   break
19: end if
20: end for
21: for  $k \in \{EA, BC\}$  do
22:    $\tau_k^* \leftarrow \arg \max_{\tau \in [0,1]} \text{Dice}(\mathbb{I}[P_{val}^k \geq \tau], Y_{val}^k; V_{val})$ 
23: end for
24: Freeze  $(\theta^*, \tau_{EA}^*, \tau_{BC}^*)$  and evaluate once on  $\mathcal{D}_{te}$ 

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Implementation safeguards. Fix random seeds, log every split and hyperparameter, use mixed precision only with gradient scaling, monitor both heads independently, and never optimize against the held-out test set.

Grad-CAM Explainability

Grad-CAM asks which shared feature-map locations most increased a chosen phenotype score. For feature map A^q and phenotype score s_k :

$$\alpha_q^k = \frac{1}{Z} \sum_{u,v} \frac{\partial s_k}{\partial A_{uv}^q}, \quad H_k = \text{ReLU} \left(\sum_q \alpha_q^k A^q \right).$$

H_{EA} and H_{BC} are computed separately and overlaid on the B-scan. A useful explanation should emphasize the predicted lesion and relevant sub-BM signal rather than borders, text, padding, or unrelated anatomy.

Grad-CAM does not prove causality. It is a model-audit tool: disagreement between the segmentation and attention map flags cases for review and may reveal shortcut learning.